

Risk-Based Chlorine Prediction under $\pm 0.1 \text{ mg L}^{-1}$ Sensor Uncertainty

Temperature and ageing-infrastructure scenarios for a Water Safety Plan

Analysis of wntr_chlorine_risk_scenariosipynb

Purpose	Convert a calibrated chlorine model and a stated sensor uncertainty into a network-wide, decision-ready risk assessment, and test that assessment against foreseeable heat and asset-condition scenarios
Model	WNTR 1.4.0 / EPANET 2.2, Net3 benchmark network (92 junctions, 117 pipes, 5 chlorine sensors)
Sensor basis	$\pm 0.1 \text{ mg L}^{-1}$ two-electrode membrane amperometric cells in flow cells; AR(1) hourly correlation $\rho = 0.40$; detection limit 0.02 mg L^{-1}
Scenarios	12 °C baseline; 16 °C warm season; 20 °C heatwave; 20 °C heatwave with ageing mains
Status	Synthetic demonstration. All limits illustrative and subject to utility approval

Summary. The analysis described here is a *sensor-conditioned probabilistic forecast* used as decision support within a Water Safety Plan (WSP). It is not a spatial measurement, not geostatistical interpolation, and not a statement that water is safe: chlorine residual is one operational indicator among the WSP's preventive barriers. It does not replace validated instruments, reference sampling, compliance monitoring, operational authority or independent surveillance.

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1 Executive summary

The notebook takes a calibrated chlorine decay model, a stated sensor specification of $\pm 0.1 \text{ mg L}^{-1}$, and an operational critical limit of 0.20 mg L^{-1} , and produces a network-wide risk assessment in the form a WSP can act on: a breach probability at every junction, a likelihood–consequence classification, a sampling plan, a verification check, and a risk register. It then re-runs that assessment under a prolonged heat episode and under ageing mains.

Six results carry the analysis.

1. **The quoted $\pm 0.1 \text{ mg L}^{-1}$ is ambiguous, and the ambiguity is consequential.** Read as one standard deviation it implies $\sigma = 0.100 \text{ mg L}^{-1}$; as a 95% interval, $\sigma = 0.051$. The calibration record in use — posterior $\text{sd}(k_w) = 0.052 \text{ m d}^{-1}$ — is internally consistent only with the *95% reading*. If the specification is in fact one standard deviation, the honest posterior is 0.087 m d^{-1} and every predictive band widens accordingly.
2. **A $\pm 0.1 \text{ mg L}^{-1}$ sensor cannot verify a 0.20 mg L^{-1} limit without a guard band.** Read as one standard deviation, a single reading must be 0.364 mg L^{-1} before the true value exceeds 0.20 mg L^{-1} with 95% confidence; readings between 0.036 and 0.364 mg L^{-1} cannot be classified either way. Correlated hourly error inflates the daily-mean standard deviation by a factor of 1.50 over the independent assumption.
3. **Sensor uncertainty reaches risk mainly through the calibration, not through today’s reading.** Re-conditioning the ensemble on a further day of observations changes breach probabilities by a mean of 0.002; the specification’s real effect is to set the width of the kinetic posterior and hence the predictive band at all 92 junctions — including the 87 without a sensor.
4. **Systematic sensor error dominates the random budget.** A $+0.05 \text{ mg L}^{-1}$ offset at the single most influential sensor — half the quoted band, and invisible against the 0.032 mg L^{-1} daily-mean random budget — moves individual breach probabilities by up to 0.119 and changes the high-risk junction count.
5. **Heat and ageing escalate risk substantially and unevenly.** Junctions above $P = 0.5$ rise $58 \rightarrow 63 \rightarrow 72 \rightarrow 86$ across the four scenarios, and demand at risk from 56% to 99%. Ageing mains at 20°C add as much as the entire $12 \rightarrow 20^\circ\text{C}$ warming, and concentrate the increase spatially. Seventeen consumer junctions carrying 44% of network demand change risk band; none is instrumented.
6. **Source dosing alone does not restore the baseline position.** A 30% inlet-dose increase under heatwave conditions reduces demand at risk from 83% to 72%, against a 56% baseline, because the first junctions to breach are long-residence dead-ends where the added dose has largely decayed on arrival.

2 Purpose, scope and interpretation

The report documents the method, mathematics and executed results of the notebook, and places the output within WSP governance. The WHO framework treats water safety planning as a catchment-to-consumer, risk-based process supported by operational monitoring, management procedures, verification and independent surveillance [7, 8]. A probabilistic chlorine forecast occupies one controlled decision-support role inside that system.

Table 2 maps the notebook’s sections onto WSP activities.

All measurements and events are synthetic. The 0.20 mg L^{-1} critical limit, the 0.30 mg L^{-1} alert level, the risk bands, the consequence terciles and the dosing options are illustrative inputs, not recommended values.

Table 1: Headline results. Probabilities are for the daily-minimum residual falling below 0.20 mg L^{-1} within the assessment window.

Analysis	Result	Meaning
Guard band, $\sigma = 0.10$	Action setpoint 0.364 mg L^{-1}	A 0.20 mg L^{-1} limit cannot be policed at a 0.20 mg L^{-1} setpoint
AR(1) inflation	$1.50\times$	Independence assumptions understate daily uncertainty by a third
Baseline risk	58/92 junctions $P > 0.5$; 56% of demand	Risk is concentrated in long-residence periphery
Systematic offset $+0.05 \text{ mg L}^{-1}$	$\max \Delta P = 0.119$	Bias control is a WSP control measure, not maintenance detail
Heatwave 20°C	72 junctions; 83% of demand	k_b multiplier 1.68 (90% range 1.44–1.95)
Heatwave + ageing mains	86 junctions; 99% of demand	Asset condition matters as much as temperature
Corrective dosing +30%	72% of demand still at risk	Dosing alone insufficient; turnover measures needed
Verification	Brier 0.015 vs climatology 0.230	Skill +0.94 at baseline; scenarios cannot be verified

Table 2: Correspondence between the analysis and WSP activities.

WSP activity	Contribution here	Required guardrail
Hazard identification and risk assessment	Breach probability and likelihood–consequence banding at every junction (§4, §8.1)	Utility-approved limits and consequence model; not a health-risk statement
Operational monitoring and critical limits	Guard-band setpoints derived from the sensor specification (§6)	Alarm ownership, persistence rules, retained instrument displays
Validation of control measures	Uncertainty budget and systematic-error stress test (§5, §9)	Reference-method uncertainty and maintenance state
Hazardous-event scenarios	Temperature and ageing scenarios with risk maps (§10)	Scenarios are planning products; they extrapolate outside calibration
Control-measure evaluation	Corrective dosing assessment (§12)	DBP, taste and acceptability constraints are not modelled
Supporting programmes	Risk-based sampling and monitoring adequacy (§13)	Field resource and access constraints
Verification	Forecast reliability and Brier skill (§14)	Independent reference sampling; does not replace compliance monitoring
Management and review	Risk register and review triggers (§15)	Version control, approval, expiry

3 Model, calibration record and observation model

3.1 Forward model

Chlorine is transported by advection with reaction, solved at network scale by EPANET using the hydraulic state. Bulk decay is first order, $r_{\text{bulk}} = k_b C$ with $k_b < 0$; the wall coefficient k_w has units of length per time and is converted by EPANET into an effective volumetric sink through pipe geometry and mass transfer. Because k_b and k_w can partially compensate, the calibration objective is neither convex nor free of ridges.

Simulations run for 48 h with hourly hydraulic and reporting steps and a 300 s water-quality step. The assessment window is hours 12–35, excluding the fill-up transient. The quantity of interest at each junction is the **daily minimum** residual within that window: a disinfection barrier is judged by its worst state, and depletion is concentrated in a few hours of the diurnal cycle.

3.2 Calibration record and its implied sensor convention

The record carried into the assessment is

$$\begin{bmatrix} k_b \\ k_w \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} -0.500 \\ -1.175 \end{bmatrix}, \begin{bmatrix} 0.050^2 & \rho \sigma_b \sigma_w \\ \rho \sigma_b \sigma_w & 0.052^2 \end{bmatrix}\right), \quad \rho = -0.288, \quad (1)$$

in d^{-1} and m d^{-1} respectively. Five sensors at nodes 243, 10, 131, 193 and 61 have nominal residence times of 36.0, 2.6, 34.6, 5.7 and 0.1 h.

The recorded $\text{sd}(k_w) = 0.052 \text{ m d}^{-1}$ is itself informative. Parametric bootstrap during calibration measured the posterior width as a function of sensor noise (Table 3); 0.052 corresponds to $\sigma \approx 0.05$, i.e. to reading the $\pm 0.1 \text{ mg L}^{-1}$ specification as a *95% interval*. The notebook makes this explicit rather than leaving it implicit, and carries the alternative readings as scenarios.

Table 3: Measured scaling of calibration precision with sensor noise (parametric bootstrap, 200 replicates per level), and the resulting predictive band width at the junction closest to the critical limit.

Sensor reading of $\pm 0.1 \text{ mg L}^{-1}$	$\sigma \text{ (mg L}^{-1}\text{)}$	$\text{sd}(\hat{k}_w) \text{ (m d}^{-1}\text{)}$	5–95% band at node 105
95% interval	0.051	0.053	0.192–0.217
one standard deviation	0.100	0.087	0.185–0.224
± 0.2 class, 1 SD	0.200	0.161	0.171–0.244

3.3 Observation model

For sensor s at hour t , $y_{s,t} = C_{s,t} + b_s + \epsilon_{s,t}$, where b_s is a standing offset and ϵ is AR(1) with hourly correlation $\rho_h = 0.40$. Readings are censored below a 0.02 mg L^{-1} detection limit. A day-mean model-discrepancy floor of 0.010 mg L^{-1} is added at each sensor, and a node-level discrepancy of $\sigma_{\text{node}} = 0.02 \text{ mg L}^{-1}$ is applied when converting predicted minima into breach probabilities.

4 Hazardous event, critical limits and consequence

The hazardous event is loss of disinfectant residual: free chlorine at a consumer node falling below the concentration the utility relies on as an operational barrier indicator. The critical limit is $C_{\text{crit}} = 0.20 \text{ mg L}^{-1}$ with an alert level at 0.30 mg L^{-1} .

A probability map is not a risk assessment. Consequence is proxied by base demand at the junction, as a surrogate for consumers exposed: total base demand is 193 L s^{-1} across 59 consumer junctions, with 33 junctions carrying no demand (hydraulic connection points, treated as negligible

consequence but retained on the maps). Consequence is banded by tercile of non-zero demand, at 1.16 and 3.39 L s⁻¹, giving 20 minor, 19 moderate and 20 major junctions. Consequence is highly concentrated: the ten largest demands carry 49% of the network total.

This is a deliberate simplification. Demand is not population, and it ignores sensitive receptors, service connections and time-of-day exposure.

5 The sensor specification and its uncertainty budget

5.1 Three readings of the same figure

A quoted $\pm 0.1 \text{ mg L}^{-1}$ is ambiguous until its coverage is stated. Read as one standard deviation, $\sigma = 0.100$; as a 95% interval, $\sigma = 0.051$; as a uniform bound, $\sigma = 0.058$. All three are carried through the analysis.

5.2 Autocorrelation

Hourly sensor errors are not independent. For AR(1) correlation ρ , the variance of a mean of n hourly values is

$$\text{Var}(\bar{\epsilon}) = \frac{\sigma^2}{n^2} \left[n + 2 \sum_{k=1}^{n-1} (n-k) \rho^k \right], \quad (2)$$

which exceeds σ^2/n . With $n = 24$ and $\rho = 0.40$, the daily-mean standard deviation is $1.50 \times$ the independent value: 0.0306 mg L^{-1} against 0.0204 mg L^{-1} at $\sigma = 0.100$. Adding the model-discrepancy floor gives a total daily-mean standard deviation of 0.0322 mg L^{-1} . Treating correlated readings as independent understates the daily uncertainty by a third, and with it every interval and alarm limit derived from it.

5.3 Random and systematic components behave differently

Random error averages down over a day of readings; a standing offset does not. For two-electrode membrane amperometric cells much of an in-service $\pm 0.1 \text{ mg L}^{-1}$ is not white noise: membrane fouling and permeability change, electrolyte consumption walking the zero point, residual temperature compensation, and the pH dependence of HOCl/OCl⁻ speciation all move the reading slowly and coherently between recalibrations. Section 9 quantifies the consequence.

6 Decision rules under measurement uncertainty

Before any modelling, the sensor specification limits what a *measurement* can settle. If a reading y carries error $\epsilon \sim \mathcal{N}(0, \sigma^2)$, then

$$P(\text{true} < C_{\text{crit}} \mid y) = \Phi\left(\frac{C_{\text{crit}} - y}{\sigma}\right), \quad (3)$$

so a reading exactly at the limit carries a 50% chance of true non-compliance. Confidence that the limit is met requires an action setpoint raised above it by a *guard band* — standard practice in conformity assessment, and rarely applied to online water-quality alarms.

The operational consequence is direct: with a 0.20 mg L^{-1} critical limit and a $\pm 0.1 \text{ mg L}^{-1}$ sensor read as one standard deviation, an alarm setpoint of 0.36 mg L^{-1} is required for 95% confidence that the limit is truly met. Averaging over a day helps, but only as far as (2) allows — and not at all against a standing offset.

7 How sensor uncertainty reaches the risk number

Two pathways exist, and they are not equally important.

Table 4: Single-reading action setpoints giving stated confidence that the true concentration exceeds 0.20 mg L^{-1} , and the resulting grey zone within which a single reading cannot be classified at 95% confidence either way.

Reading of $\pm 0.1 \text{ mg L}^{-1}$	σ	90% setpoint	95% setpoint	95% grey zone
95% interval	0.051	0.265	0.284	0.116–0.284 (width 0.168)
one standard deviation	0.100	0.328	0.364	0.036–0.364 (width 0.329)
uniform bound	0.058	0.274	0.295	—

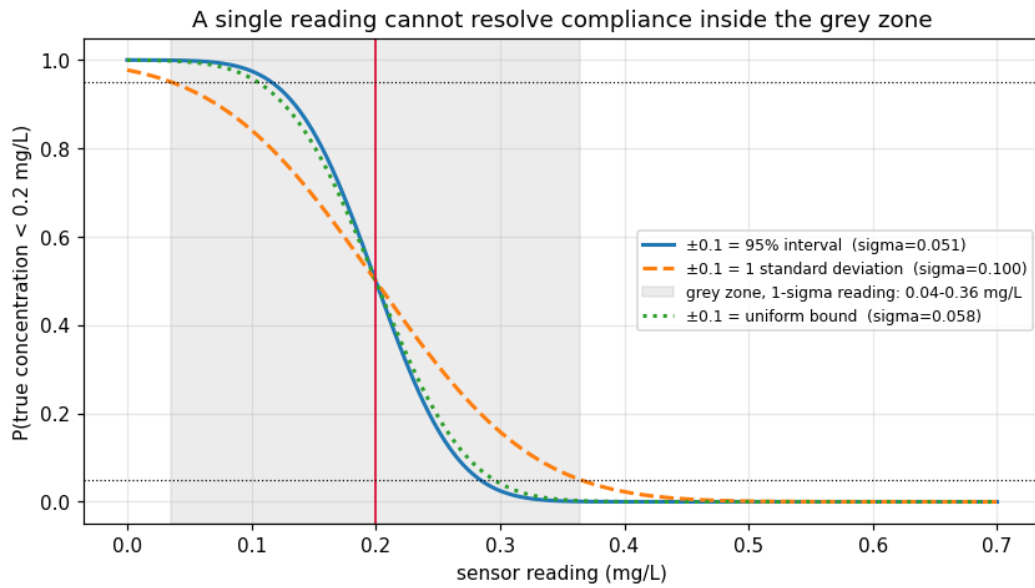


Figure 1: Probability that the true concentration is below 0.20 mg L^{-1} as a function of the sensor reading, under the three readings of the specification. The shaded band is the grey zone for the one-standard-deviation case: readings inside it cannot be classified at 95% confidence in either direction.

Pathway 1: through the calibration (dominant). The wall coefficient is estimated from the same sensors, so a noisier sensor yields a wider posterior which propagates into a wider predictive distribution at every junction. Table 3 gives the measured scaling; the predictive band at the junction closest to the limit widens from 0.025 to 0.073 mg L⁻¹ across the three readings, approximately linearly in σ as Fisher information requires.

Pathway 2: through the live reading. This is the guard-band result of §6, independent of the model.

What conditioning on today’s record adds: very little. The ensemble is importance weighted against the current day’s sensor day-means. The effective sample size remains 207–216 of 220 draws, and the mean change in breach probability is 0.002. A single further day of observation cannot materially update a posterior already built from a multi-day calibration. The product is therefore properly described as a **calibration-conditioned forecast with a same-day consistency check**, not a nowcast. Reporting this honestly matters: the term “sensor-conditioned” should not be read as implying that the map tracks the sensors hour by hour.

8 Baseline risk assessment

Under current conditions (12°C, ± 0.1 mg L⁻¹ read as one standard deviation), 58 of 92 junctions have $P > 0.5$, carrying 56% of network demand. Eleven junctions fall in the indeterminate band $0.05 < P < 0.95$, i.e. the assessment cannot classify them either way.

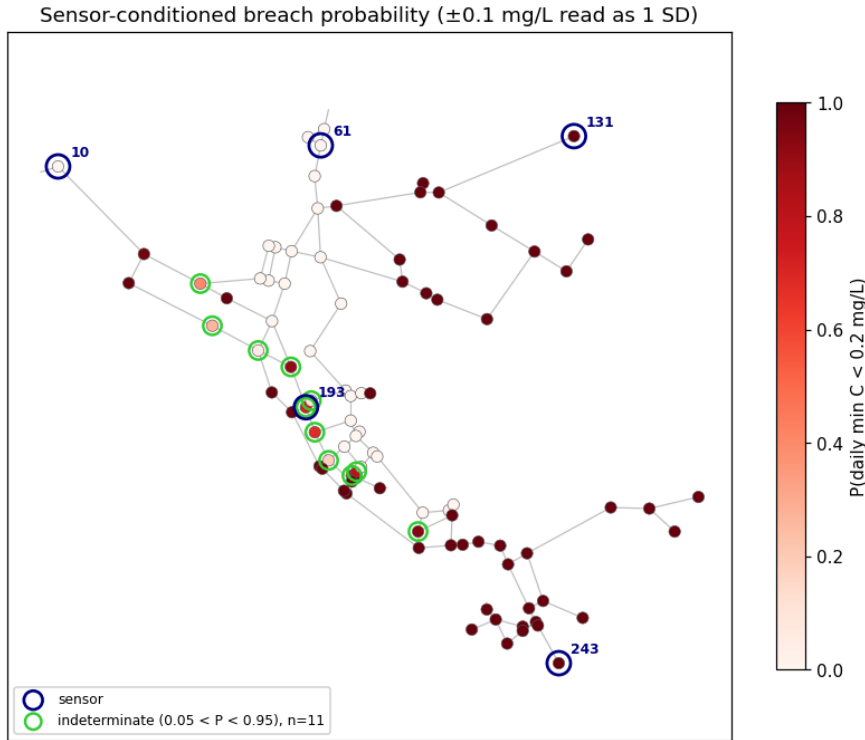


Figure 2: Baseline breach probability. Navy circles mark the five chlorine sensors; green rings mark the eleven junctions whose classification is unresolved. The unresolved cluster sits around sensor 193, in the transition zone between the well-supplied core and the depleted periphery.

The effect of the sensor specification on the *bulk* classification is small: no junction crosses $P = 0.5$ between readings, because most of this network is either comfortably above the limit or fully depleted. The cost appears in the marginal band, where junctions classified at 95% confidence

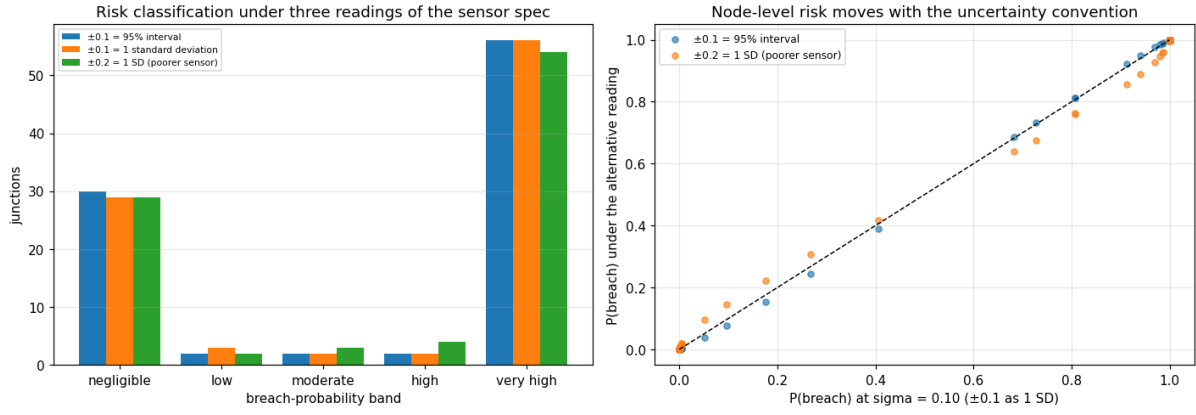


Figure 3: Left: distribution of junctions across probability bands under the three readings of the sensor specification. Right: node-level breach probability under alternative readings against the one-standard-deviation case.

fall from 82 to 81 to 79 of 92 as the specification loosens. Those marginal junctions are precisely where sampling effort and operational decisions are directed.

8.1 Risk classification

Likelihood bands (rare, unlikely, possible, likely, almost certain) are combined with the four consequence classes to give a risk score and band. The result is 26 very-high, 2 high, 17 medium and 14 low junctions, with 33 not applicable (no demand). The 28 junctions in the high and very-high bands carry 122.8 L s^{-1} , or 64% of network demand.

9 Systematic sensor error: stress test

Sensor influence on the assessment is highly uneven. Measured as the ensemble spread of the predicted day-mean, node 193 (residence 5.7 h) dominates at 0.0147 mg L^{-1} , against 0.0040 at node 10, 0.0018 at node 131, 0.0008 at node 61 and 0.0000 at node 243. The most depleted sensor carries no information about the kinetics — a result that reproduces, without prompting, the intermediate-residence information peak established earlier in the project.

Table 5: Effect of a standing offset on the risk assessment. The random daily-mean budget is 0.0322 mg L^{-1} ; none of these offsets would be detected by it.

Scenario	ESS	$P > 0.5$ junctions	mean $ \Delta P $	max $ \Delta P $	high/very-high
No bias	213.9	58	—	—	28
+0.05 mg L^{-1} at node 193	136.3	58	0.010	0.119	27
−0.05 mg L^{-1} at node 193	181.8	59	0.009	0.116	29
+0.05 mg L^{-1} at all sensors	124.3	58	0.011	0.124	27

A half-band offset at a single sensor is invisible against the random budget yet moves individual probabilities by up to 0.119 and changes the high-risk count. Bias control — reference checks, recalibration cadence, fouling management — is therefore a WSP control measure with a quantified effect on the risk assessment, not a maintenance housekeeping item.

10 Hazardous-event scenarios: prolonged heat and ageing mains

10.1 Temperature dependence

A risk assessment fixed at today’s conditions is of limited planning use. The event modelled is a prolonged 30 °C air-temperature episode raising in-pipe water temperature from about 12 to about 20 °C, in a network with a proportion of old mains.

Both decay pathways follow Arrhenius behaviour to good approximation over this range:

$$k(T) = k(T_{\text{ref}}) \exp \left[-\frac{E_a}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right], \quad T \text{ in K.} \quad (4)$$

Reported activation energies for bulk chlorine decay cluster around 30–60 kJ mol^{−1} [6, 2, 4], roughly a doubling of rate per 10 °C. Wall reactions are also temperature-sensitive, and in older mains the wall term includes corrosion product and biofilm whose activity responds strongly to warming [1].

Three quantities are uncertain simultaneously and all are sampled: the kinetic coefficients (width set by the sensor specification); the activation energies, $E_{a,\text{bulk}} \sim \mathcal{N}(45, 8^2)$ and $E_{a,\text{wall}} \sim \mathcal{N}(35, 10^2)$ kJ mol^{−1}; and the water temperature reached, $\mathcal{N}(T_{\text{scen}}, 1^2)$ °C.

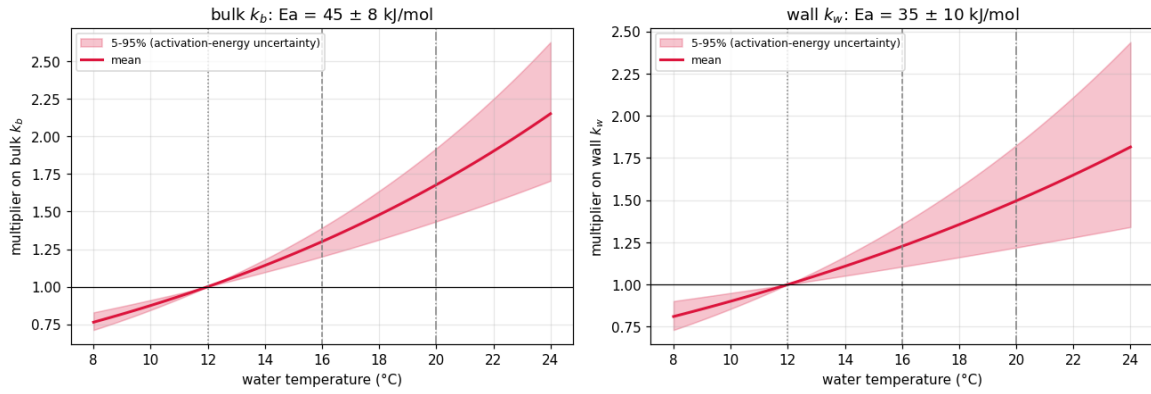


Figure 4: Temperature multipliers on the bulk and wall coefficients relative to the 12 °C calibration reference, with the band induced by activation-energy uncertainty. Vertical lines mark the three scenario temperatures.

At 20 °C the bulk multiplier is 1.68 with a 90% range of 1.44–1.95, and the wall multiplier 1.50; the implied coefficients are $k_b = -0.839 \text{ d}^{-1}$ and $k_w = -1.758 \text{ m d}^{-1}$. The activation energy is therefore a material source of scenario uncertainty in its own right, not a fixed constant.

10.2 Scenario definitions

Scenario D additionally applies age-weighted wall factors by pipe group, from 0.70 for the newest to 1.85 for the oldest, representing both higher baseline wall reactivity and a stronger response to warming in deteriorated mains.

Table 6: Scenario results. All at $\pm 0.1 \text{ mg L}^{-1}$ read as one standard deviation.

Scenario	$\bar{k}_b$ (d ^{−1})	$\bar{k}_w$ (m d ^{−1})	$P > 0.5$ junctions	Demand at risk (L s ^{−1})	% demand	High/v.high risk	Indetermin
A. Baseline 12 °C	−0.507	−1.186	58	107.9	56%	29	11
B. Warm season 16 °C	−0.653	−1.464	63	146.8	76%	31	17
C. Heatwave 20 °C	−0.849	−1.771	72	160.5	83%	34	24
D. Heatwave + ageing mains	−0.854	−1.759	86	189.9	99%	38	6

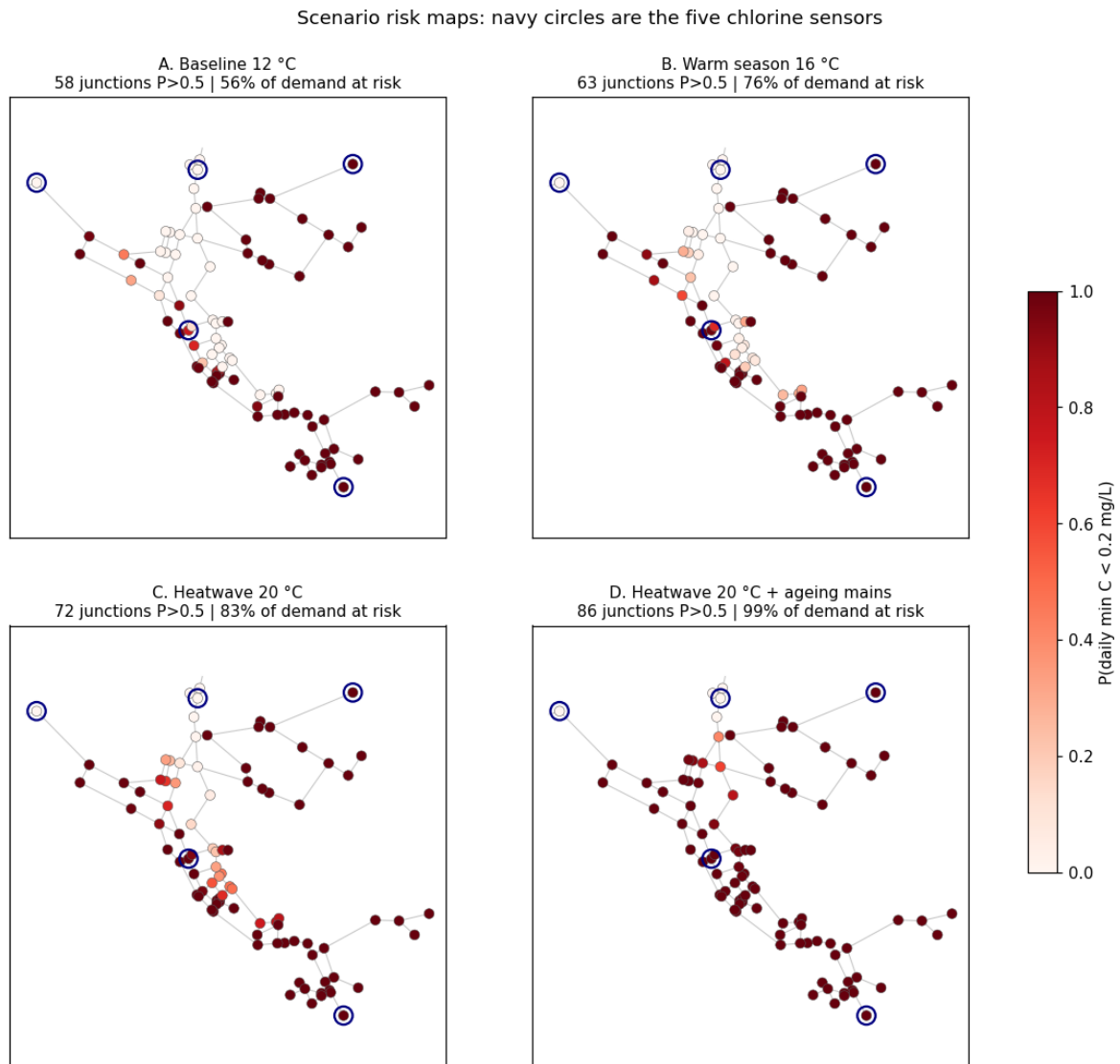


Figure 5: Scenario risk maps. Navy circles mark the five sensors. The progression is not uniform: warming deepens risk in the already-marginal periphery, while ageing mains extend it into the central corridor.

10.3 Reading the escalation

Junctions above $P = 0.5$ rise from 58 at baseline to 72 under heatwave (+14), and ageing mains at the same temperature add a further 14, to 86. Demand at risk rises from 56% to 99%. Two features deserve comment.

First, **asset condition matters as much as temperature**: the ageing increment equals the entire $12 \rightarrow 20^\circ\text{C}$ warming increment, which argues for treating mains condition as a first-order input to chlorine risk rather than a secondary correction.

Second, the indeterminate count behaves non-monotonically — $11 \rightarrow 17 \rightarrow 24 \rightarrow 6$. Warming moves junctions into the marginal band, but the ageing scenario pushes most of them decisively past it. Scenario D is not a better-resolved assessment; it is a worse outcome about which the model happens to be more certain.

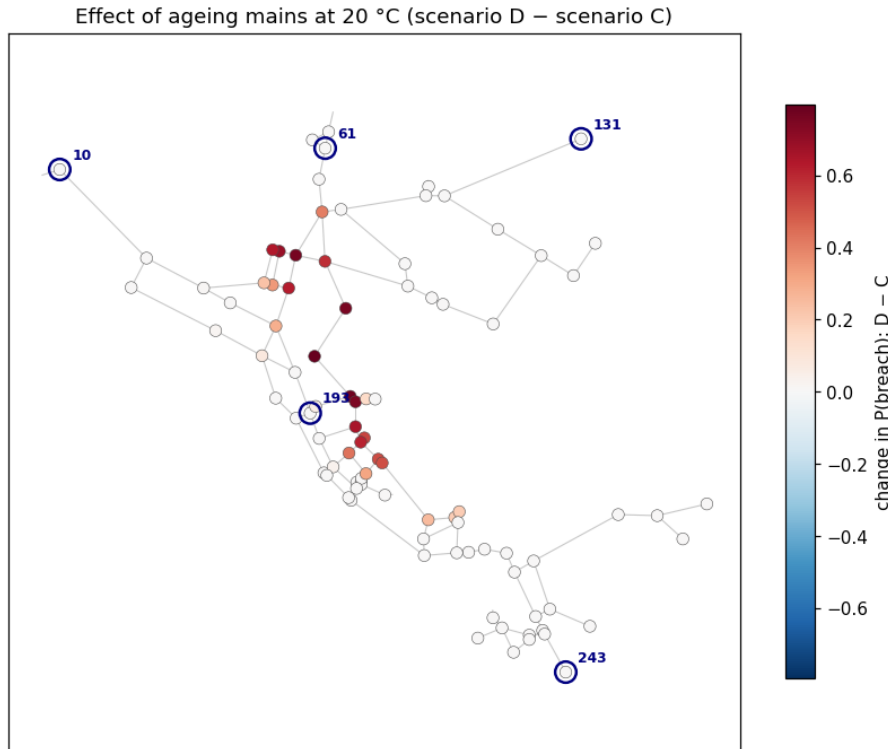


Figure 6: Change in breach probability attributable to ageing mains at constant temperature (scenario D minus scenario C). The increase is spatially concentrated rather than uniform.

The largest ageing-driven increases are +0.795 at node 159, +0.771 at node 161, +0.762 at node 120, +0.753 at node 157 and +0.746 at node 163, with a net change in demand at risk of $+29.4 \text{ L s}^{-1}$.

11 Compounded uncertainty: sensor specification and scenario

The two uncertainty sources interact. A heat episode pushes more junctions toward the critical limit, and it is near the limit that sensor uncertainty determines whether a junction can be classified at all. Repeating the heatwave scenario under each reading of the specification gives 70, 68 and 68 junctions classified at 95% confidence for $\sigma = 0.051$, 0.100 and 0.200 respectively.

The larger effect is the scenario itself: indeterminate junctions rise from 11 at baseline to 24 under heatwave at the same specification. A heat episode therefore degrades the assessment twice over — it raises the risk, and it moves more of the network into the band where the existing five sensors cannot resolve the classification. Monitoring adequacy is not a fixed property of an array; it is conditional on the operating regime.

12 Control measure: corrective dosing

Under first-order kinetics the residual scales approximately linearly with inlet concentration, making source dosing the obvious operational response.

Table 7: Corrective dosing under heatwave conditions. Baseline demand at risk (12°C , 1.0 mg L^{-1} inlet) is 107.9 L s^{-1} .

Inlet dose (mg L^{-1})	$P > 0.5$ junctions	Demand at risk (L s^{-1})	% of demand	99.5th pct. node residual
1.00	71	160.5	83%	0.98
1.15	66	149.6	78%	1.12
1.30	62	137.8	72%	1.27

A 30% inlet-dose increase leaves 137.8 L s^{-1} at risk against 107.9 L s^{-1} at baseline: **source dosing alone does not restore the pre-heatwave risk position on this network**. The reason is structural. The junctions that breach first are long-residence dead-ends, where the additional dose has largely decayed before arrival; raising the source concentration lifts the whole profile but cannot buy back residence time. Turnover measures — flushing, storage management, rezoning — act directly on the controlling variable and should be evaluated alongside dosing rather than after it.

This is a control-measure evaluation, not a dosing recommendation. Raising the dose trades residual risk against disinfection by-product formation, taste and consumer acceptability, none of which is modelled, and the required increase is itself uncertain because it is computed from the same temperature-extrapolated kinetics.

13 Risk-based sampling and monitoring adequacy

Sampling priority is not simply the highest probability: a junction already known to breach with near-certainty yields little new information. The informative locations are those whose classification is uncertain and whose consequence is high, so priority is scored by consequence weighted by classification uncertainty $P(1 - P)$, which peaks at $P = 0.5$.

The five highest priorities are nodes 105 ($P = 0.406$, 8.5 L s^{-1}), 193 (0.727, 4.5), 109 (0.268, 14.6), 189 (0.175, 6.8) and 113 (0.913, 1.3). Of the top ten, **only one is currently instrumented**; nine require manual confirmatory sampling. Eleven junctions remain unresolved, six of which serve moderate or major demand.

Five sensors observe 92 junctions directly. Every unmonitored probability is model-mediated and inherits the structural, hydraulic and boundary assumptions of §16.

14 Verification

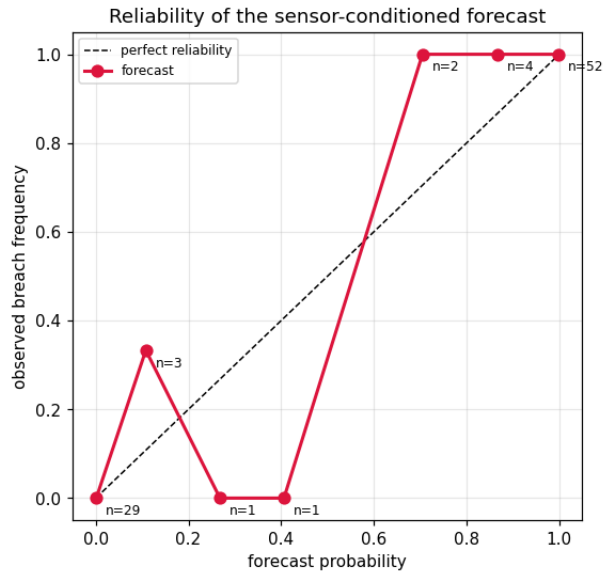
Because the synthetic reality is known, the baseline forecast can be verified directly: junctions are grouped by forecast probability and compared with the frequency of actual breach. In reality 59 of 92 junctions breach the limit within the assessment window.

The Brier score is 0.015 against a climatological 0.230, a skill of +0.94. The extremes are well calibrated; the middle bands are sparsely populated and noisy. Residual miscalibration is attributable to structural error — the forecast uses a homogeneous wall-decay model while the synthetic reality is heterogeneous — which parameter uncertainty alone cannot represent. This is precisely why verification is a separate WSP activity rather than a by-product of calibration.

The scenario maps cannot be verified in this way. They are projections into a regime for which no observations exist, and their credibility rests on the Arrhenius extrapolation rather than on evidence.

Table 8: Reliability of the baseline forecast.

Forecast band	n	Mean forecast P	Observed frequency
0.00–0.05	29	0.001	0.000
0.05–0.20	3	0.108	0.333
0.20–0.40	1	0.268	0.000
0.40–0.60	1	0.406	0.000
0.60–0.80	2	0.705	1.000
0.80–0.95	4	0.867	1.000
0.95–1.00	52	0.998	1.000

**Figure 7:** Reliability diagram for the baseline forecast. Bin counts are annotated; the middle bands contain 1–4 junctions and are correspondingly noisy.

15 Risk register and review triggers

The register is generated from the analysis rather than transcribed, so it remains consistent with the model version and sensor assumptions. It carries current and scenario probabilities side by side, so that locations which escalate under heat are visible before the event.

Seventeen consumer junctions change risk band between current conditions and scenario D, carrying 84.6 L s^{-1} or 44% of network demand. Table 9 lists the most significant. **None of them is instrumented**, and several move from “low” to “very high”: node 119, the largest single demand at 11.1 L s^{-1} , moves from $P = 0.000$ to $P = 0.608$.

Table 9: Consumer junctions escalating under the heat and ageing scenario (extract).

Node	P current	P heatwave	P heat + ageing	Demand (L s^{-1})	Band current	Band scenario D
105	0.406	0.989	1.000	8.54	high	very high
109	0.268	0.982	1.000	14.60	high	very high
189	0.175	0.967	1.000	6.81	medium	very high
111	0.052	0.914	1.000	8.96	medium	very high
199	0.002	0.746	1.000	7.53	low	very high
117	0.000	0.358	0.984	7.43	low	very high
119	0.000	0.018	0.608	11.11	low	very high

The assessment carries explicit expiry conditions: a network surveillance flag; sensor QA failure, disagreement with a reference check, or sensor service; revision of the hydraulic model or demand allocation; a change in source or treatment regime; water temperature moving outside the $8\text{--}24^\circ\text{C}$ scenario range; a forecast heat episode (switch to the scenario D register and pre-authorise dosing); mains rehabilitation or a burst altering the assumed age profile; or the calibration record exceeding its approved age.

16 Critical review and limitations

What the analysis establishes. That a stated sensor specification can be carried coherently through calibration, prediction, risk classification, scenario testing and control evaluation, and that doing so exposes decisions — guard-band setpoints, sampling locations, the insufficiency of dosing — that a deterministic treatment would leave hidden.

What it does not. The following limitations are material and are stated in the notebook itself.

- **Sparse observability.** Five sensors, 92 junctions. Every unmonitored probability is model-mediated, not measured.
- **Structural error is outside the uncertainty budget.** The ensemble varies two kinetic parameters around a homogeneous model while reality is heterogeneous. Section 14 shows the resulting reliability gap. Parameter uncertainty is a floor on total uncertainty, not an estimate of it.
- **Scenarios extrapolate outside the calibrated regime.** The calibrated coefficient is an *effective* parameter fitted at one temperature and asset condition. Scaling it by a laboratory Arrhenius law assumes the aggregated system inherits the temperature response of the underlying reactions. Activation-energy uncertainty is sampled, but the *form* of the extrapolation is an assumption.
- **Ageing factors are illustrative.** Pipe groups are length quintiles, a placeholder for material and installation-year records; the 0.70–1.85 multipliers are plausible magnitudes, not measurements for this network.

- **Temperature is applied uniformly and instantaneously.** A real heat episode warms the network unevenly and with lag, and would also change demand, turnover and residence time, none of which is represented.
- **Hydraulics, demands and boundary conditions are treated as known.**
- **The consequence model is a proxy.** Base demand is not population and ignores sensitive receptors.
- **The sensor error model is simplified.** A Gaussian AR(1) with a single ρ does not capture fouling episodes, temperature and flow sensitivity, lag or maintenance-dependent bias; the stress test probes only a static offset.
- **Synthetic throughout.** Results demonstrate method behaviour and establish nothing about performance on a real network.

Recommended next steps. In order of value: represent hydraulic and boundary uncertainty in the ensemble; add a structural-discrepancy term informed by the verification gap, so that predictive intervals are not merely parameter intervals; replace the length-quintile age proxy with asset records before the ageing scenario is used for planning; determine the temperature response empirically from seasonal operational data rather than assuming it; and evaluate turnover measures alongside dosing in the control-measure assessment.

17 Conclusions

The analysis converts a sensor specification into decisions rather than into error bars. Three conclusions transfer beyond the synthetic case.

First, **a $\pm 0.1 \text{ mg L}^{-1}$ specification is not adequate to police a 0.20 mg L^{-1} limit at face value.** Either the alarm setpoint is guard-banded upward — to 0.36 mg L^{-1} for 95% confidence under the conservative reading — or the utility accepts a wide grey zone in which compliance is genuinely undetermined. The convention attached to the $\pm$ figure must be recorded, because it changes the answer.

Second, **the error that matters is the one that does not average away.** Random error is handled by the daily mean; a half-band standing offset at one influential sensor is invisible to that budget yet moves the risk classification. Sensor QA is a control measure with quantified consequences.

Third, **climate and asset condition are of comparable importance, and neither is addressed by dosing alone.** Warming from 12 to 20°C and ageing mains at 20°C each add fourteen junctions above $P = 0.5$; together they place 99% of network demand at risk, and a 30% dose increase recovers only part of it. The junctions that escalate are unmonitored, which is an argument for information-led monitoring design rather than for more monitoring in general.

The correct description of the product is a versioned, sensor-conditioned probabilistic forecast with stated expiry conditions. Within a WSP it can strengthen hazard assessment, sampling design, incident investigation and improvement planning. It remains subordinate to the plan's preventive controls, competent operational judgement, validated instruments, reference testing, compliance monitoring and independent surveillance.

A Reproducibility record

The notebook contains 37 cells, of which 21 are code cells; all carry execution counts from a single top-to-bottom pass, none produced an error output, and seven figures are embedded. The recorded environment is Python 3.10 with WNTR 1.4.0 driving EPANET 2.2. Ensembles use 220 draws for the sensor-specification comparison and 150 per temperature scenario, with 90 draws

per dosing option; random seeds are fixed per cell so results are reproducible independently of execution order. Exact numerical reproduction additionally depends on the EPANET engine build, the bundled Net3 input file, and platform floating-point behaviour. The generated risk register is written to `chlorine_risk_register.csv` (92 rows, current and scenario columns).

B Notation

Symbol	Meaning
C	free chlorine concentration (mg L^{-1})
C_{crit}	operational critical limit, 0.20 mg L^{-1}
k_b	first-order bulk decay coefficient (d^{-1})
k_w	wall decay coefficient (m d^{-1})
σ	sensor error standard deviation (mg L^{-1})
ρ	AR(1) hourly correlation of sensor error
σ_{node}	node-level model-discrepancy standard deviation
E_a	Arrhenius activation energy (kJ mol^{-1})
P	probability that the daily-minimum residual falls below C_{crit}
ESS	importance-weight effective sample size

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